

The Johnson Family $J(N, 1)$ Has No Total-Variation Cutoff

Conjecture 00000003961

Abstract

The assertion of total-variation cutoff for all $k \leq N/2$ is false. For the valid family $k = 1$, the Johnson graph is K_N . Its standard half-lazy random walk has $t_{\text{mix}}(1/4) = 2$ and $t_{\text{mix}}(3/4) = 1$ for every $N \geq 5$. Their ratio is constantly 2, contrary to the cutoff criterion that it tend to 1. We prove the exact transition and total-variation formulas, not merely an asymptotic estimate.

Definitions and the claimed range

The vertices of $J(N, 1)$ are the singleton subsets of $\{1, \dots, N\}$. Distinct singletons have symmetric difference of size 2, so $i \mapsto \{i\}$ identifies this actual Johnson graph with the complete graph K_N . We use the standard lazy walk: stay at the current vertex with probability $1/2$, and otherwise choose uniformly among its $N - 1$ neighbors. Thus

$$P(i, j) = \begin{cases} 1/2, & i = j, \\ 1/[2(N - 1)], & i \neq j. \end{cases} \quad \pi(j) = \frac{1}{N}.$$

The matrix is positive and doubly stochastic, so the uniform distribution π is stationary. In particular, there is no disconnectedness or periodicity obstruction in this example.

Write

$$d_N(t) = \max_i \frac{1}{2} \sum_j |P^t(i, j) - \pi(j)|, \quad t_{\text{mix}}^{(N)}(\varepsilon) = \min\{t \geq 0 : d_N(t) \leq \varepsilon\}.$$

The standard total-variation cutoff criterion is that, for every $0 < \varepsilon < 1/2$,

$$\frac{t_{\text{mix}}^{(N)}(\varepsilon)}{t_{\text{mix}}^{(N)}(1 - \varepsilon)} \longrightarrow 1 \quad (N \longrightarrow \infty). \quad (1)$$

See [1] for this formulation. The source quantifies over all $k \leq N/2$ and does not require $k \rightarrow \infty$. Hence the family $k = 1$, $N \geq 5$ is within its stated range.

Proposition 1. *For the half-lazy walk on $J(N, 1)$, $N \geq 5$, set $a_N = (N - 2)/[2(N - 1)]$. For every nonnegative integer t ,*

$$d_N(t) = \frac{N - 1}{N} a_N^t. \quad (2)$$

In particular, $t_{\text{mix}}^{(N)}(1/4) = 2$ and $t_{\text{mix}}^{(N)}(3/4) = 1$.

Proof. Let U be the $N \times N$ matrix all of whose entries equal $1/N$. Direct multiplication gives $U^2 = U$. The transition probabilities and induction then give

$$P = a_N I + (1 - a_N)U, \quad P^t = a_N^t I + (1 - a_N^t)U.$$

For a fixed starting state i , subtracting π gives the diagonal difference $a_N^t(1 - 1/N)$ and each of the other $N - 1$ differences $-a_N^t/N$. Since $a_N \geq 0$, their absolute values sum to $2(N - 1)a_N^t/N$. This proves (2); it is independent of i , so the maximum over initial states is exactly the same value. Also $a_N < 1$, so these distances tend to zero for each fixed N and every positive tolerance has a finite mixing time.

At the first three times, the formula becomes

$$d_N(0) = \frac{N - 1}{N} > \frac{3}{4}, \quad d_N(1) = \frac{N - 2}{2N}, \quad d_N(2) = \frac{(N - 2)^2}{4N(N - 1)}.$$

For $N \geq 5$, $d_N(1) > 1/4$ because $N > 4$, while $d_N(1) < 1/2 < 3/4$. Furthermore,

$$(N - 2)^2 \leq N(N - 1) \iff 4 \leq 3N,$$

so $d_N(2) \leq 1/4$. These comparisons show both the upper bounds and the minimality of the asserted mixing times. $\square$

Taking $\varepsilon = 1/4$ in (1) now yields the constant ratio $2/1 = 2$, whose limit is not 1. Thus this family has no total-variation cutoff. This already refutes the universal cutoff clause and hence the conjecture as written; no analysis of its proposed cutoff-time multiplier is needed. This elementary complete-graph calculation is not claimed as a new result about Markov chains. The conclusion does not address a different statement restricted to growing k .

Formal verification and reproduction

The accompanying Lean project uses Lean 4.19.0 and Mathlib at commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

It indexes $N \geq 5$ by $N = n + 5$. The actual singleton-subset graph, bijective labeling, adjacency, and neighbor count are verified. The transition matrix is matched to that graph. Mathlib's `doublyStochastic` structure verifies its probability entries and every matrix power, and the uniform distribution is proved stationary.

The matrix-power identity, exact finite total-variation sum, convergence, and existence of mixing times are formalized. Mixing time is the infimum of the actual nonempty set of admissible integer times; its least values 2 and 1 are proved. Universal quantification over starting states implements the maximum bound. The final theorem `no_total_variation_cutoff` contradicts the real-limit criterion (1). No spectral-gap assertion is assumed.

The exact repository statement is in `SOURCE.md`; build commands and detailed correspondence are in `README.md`. Dependencies use public, pinned Git revisions. The verification directory records actual fresh builds, axiom audits, hashes, and PDF checks. The auxiliary `verify.py` constructs the subset graphs and multiplies rational matrices at five finite sizes. It supplements the general Lean proof.

References

- [1] J. Hermon and Y. Peres, *On sensitivity of mixing times and cutoff*, Electronic Journal of Probability **23** (2018), paper 25. arXiv:1610.04357. The cutoff ratio criterion appears in the abstract; the counterexample calculation here is self-contained.